

SYSTEMATIC REVIEW

Systematic review: What is the impact of triage implementation on clinical outcomes and process measures in low- and middle-income country emergency departments?

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Abstract

Introduction: Triage is widely regarded as an essential function of emergency care (EC) systems, especially in resource-limited settings. Through a systematic search and review of the literature, we investigated the effect of triage implementation on clinical outcomes and process measures in low- and middle-income country (LMIC) emergency departments (EDs).

Methods: Structured searches were conducted using MEDLINE, CENTRAL, EMBASE, CINAHL, and Global Health. Eligible articles identified through screening and full-text review underwent risk-of-bias assessment using the Newcastle-Ottawa Scale. The quality of evidence for each effect measure was summarized using GRADE.

Results: Among 10,394 articles identified through the search strategy, 58 underwent full-text review and 16 were included in the final synthesis. All utilized pre-/postintervention methods and a majority were single center. Effect measures included mortality, waiting time, length of stay, admission rate, and patient satisfaction. Of these, ED mortality and time to clinician assessment were evaluated most frequently. The majority of studies using these outcomes identified a positive effect, namely a reduction in deaths and waiting time among patients presenting for EC. The quality of the evidence was moderate for these measures but low or very low for all other outcomes and process indicators.

Conclusions: There is moderate quality of evidence supporting an association between the introduction of triage and a reduction in deaths and waiting time. Although the available data support the value of triage in LMIC EDs, the risk of confounding and publication bias is significant. Future studies will benefit from more rigorous research methods.

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INTRODUCTION

Emergency care (EC) is an integrated platform for the delivery of time-sensitive health care services, and a critical tool for achieving universal health coverage.^{1,2} It is estimated that 54% of all deaths in low- and middle-income countries (LMICs) could be addressed through effective EC, with more than one billion disability-adjusted life-years saved each year.^{3,4}

Facilities that provide EC, including emergency departments (EDs), require systems for identifying and prioritizing patients with urgent care needs. This process, referred to as triage, is relevant wherever demands for care exceed the available resources. Triage is recognized as a core function of EC facilities in the World Health Organization (WHO)'s Emergency Care Systems Framework⁵ and a central component of the Essential Emergency and Critical Care package.⁶

Triage approaches used in high-resource environments do not necessarily translate to LMICs.⁷ A small number of tools have been specifically developed for LMIC contexts, including the South African Triage Scale (SATS), Emergency Triage Assessment and Treatment (ETAT), and Interagency Integrated Triage Tool (IITT).^{8–10} These instruments have demonstrated acceptable reliability and validity in a variety of EC facilities.^{11–20} Triage tools also appear to serve additional functions in LMIC settings, such as facilitating patient flow and ED structure.^{7,17,21,22}

Although there is widespread recognition of the potential value of triage in LMIC EDs,^{10,19,20,23–26} there are limited data regarding the association between the introduction of triage and enhanced quality of care. To our knowledge, no systematic search and review has ever been conducted on this topic. Previous reviews of triage systems in LMICs have focused on the validity and reliability of individual scales, but not whether the implementation of such tools has the potential to improve the quality and timeliness of care.^{19,20} This review aimed to address this knowledge gap by determining the impact of triage introduction on clinical outcomes and process measures in LMIC EDs.

METHODS

We performed a systematic search and review of the literature to identify articles meeting prespecified inclusion criteria. Detailed study methods are described in the review protocol, which was registered and published on PROSPERO (CRD42023394402).

Structured searches were conducted in MEDLINE, EMBASE, CENTRAL (Cochrane Library), CINAHL, and Global Health using subject headings, free-text terms, and synonyms for “low- and middle-income countries” (concept A), “triage” (concept B), and “emergency department” (concept C) for the years 1990 until the present. This period was chosen as it reflects the time frame over which recognized triage systems have generally been studied. The complete search strategy, as executed on February 20, 2023, is available in Appendix S1.

To be included, studies needed to describe the introduction of a triage system in an LMIC EC facility-(ies) and report a clinical outcome or process measure associated with implementation of the system. Exclusion criteria were prospectively defined (Appendix S2). In the interest of alignment with other research on this topic, the search strategy, along with the inclusion and exclusion criteria, were informed by a recent review of triage performance in LMICs.¹⁹

Review process

Two reviewers independently screened the titles and abstracts of all search results. If selection criteria were met based on title or abstract, or it was unclear whether the study should be included, the full text was retrieved. Disagreements were resolved by discussion. If a consensus decision could not be achieved, the article was considered potentially relevant and the full text was sought. The authors of conference abstracts were contacted to provide full text of unpublished manuscripts, if an email address was provided. Screening used blind assessment methods and was performed using Covidence (Melbourne, Australia).

The full text of potentially relevant articles was independently screened by two reviewers (RM and QWT). The same consensus process was used to determine which studies were included. In cases of persisting disagreement, the input of a third reviewer (GOR) was sought. The agreement between reviewers at each stage of the screening process was assessed using the kappa statistic, as generated by Covidence.

Data extraction and classification

For studies meeting selection criteria, relevant data were independently entered by two extractors into a custom-designed data collection sheet embedded in Covidence. Differences were resolved by rereview of the original manuscript. Effect measures were categorized into outcomes and processes based on the widely utilized framework described by Donabedian.^{27,28}

Risk of bias within individual studies

The risk of bias for each included study was assessed using the Newcastle-Ottawa Scale (NOS).²⁹ The NOS is a widely used tool for assessing the quality of nonrandomized studies, which were anticipated to predominate. Application of the scale has been described in detail elsewhere.²⁹ In brief, it assesses the risk of bias across three domains (selection, comparability, and outcomes) and awards each study a score out of nine.

Assessment of individual studies using the NOS was undertaken using a dedicated assessment form embedded in Covidence. Studies with NOS scores of 0–3, 4–6, and 7–9 were considered at high risk

of bias, moderate risk of bias, and low risk of bias, respectively. Qualitative studies, if identified, were to be appraised using the Critical Appraisal Skills Programme tool for qualitative research.³⁰

Appraisal of the body of evidence

To assess the quality of evidence at outcomes level, the GRADE approach was used. GRADE evaluates the body of evidence as a whole and allocates a quality grading based on the design, methods, and results of the constituent studies.³¹ Using this approach, observational studies are initially defined as low-quality evidence, whereas randomized controlled trials are high-quality evidence. The quality assessment for each outcome measure may be adjusted depending on the nature of the studies and their findings. For instance, upgrading occurs if there is a large effect size, and downgrading occurs if there is inconsistency between studies or a high likelihood of publication bias.³¹

Reporting and synthesis of results

Review outcomes are reported here in accordance with Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines.³² We anticipated a large amount of heterogeneity among the included studies and did not expect that it would be appropriate to perform a meta-analysis. Rather, we have presented the results in narrative and tabular form.

RESULTS

Search results

Our search identified 10,394 articles, of which 7145 were unique and underwent title and abstract screening. Results are summarized in the PRISMA diagram at [Figure 1](#). Inter-rater agreement for the two primary reviewers at the title and abstract screening stage (RM and WF) was 0.68, demonstrating substantial agreement. A similar result was obtained for full-text review ($\kappa=70.0$), which was performed by RM and QWT.

Among the 58 full-text articles retrieved, 16 were included in the final synthesis ([Table 1](#)). All were quantitative in nature. Details of excluded studies, along with the reason for exclusion, are summarized in [Table 2](#). Several studies focused on the implementation of the Emergency Triage Assessment and Treatment Plus Admission program were excluded, because that package incorporates a range of simultaneous process changes and quality improvement interventions, including clinical training and guidelines development, with a focus on early inpatient care.

Summary of included studies

The 16 included studies provided data on more than 900,000 participants ([Table 1](#)). Ten were focused exclusively on pediatric populations. All studies used pre-post intervention methods, variably defined as “quasi-experimental,” cohort, or observational studies.

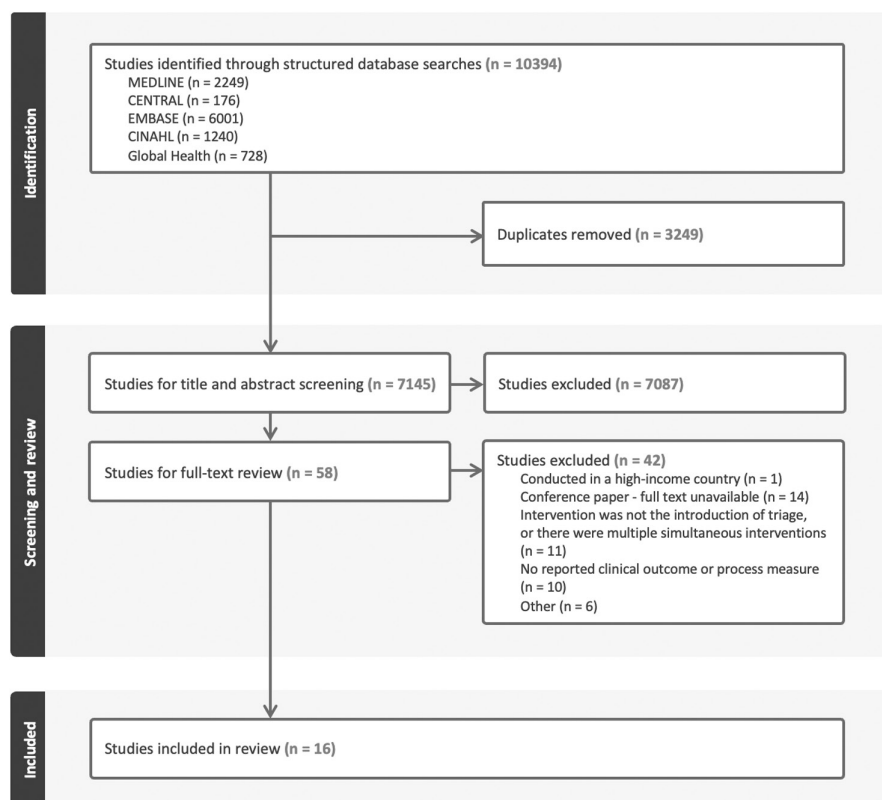


FIGURE 1 PRISMA diagram.

TABLE 1 Studies included in quantitative synthesis.

First author and year	Start date	End date	Triage system ^a	Facility type ^b	Population	Country	Country income status	Total participants (n)	Effect measures	Risk of bias ^c
Bruijns 2008 ¹²	December 2004	March 2005	CTS (SATS)	Hospital	Adult and pediatric	South Africa	Upper-middle income	1146	Waiting time (overall and by triage category)	Moderate
Cai 2020 ³³	January 2012	December 2018	Two step, four level	Hospital	Pediatric	China	Upper-middle income	645,473	In-hospital mortality; unexpected resuscitation rate; complaints about quality; complaints about ED congestion; patient satisfaction; LOS	Moderate
Crouse 2016 ³⁴	November 2009	October 2011	CETEP (ETAT)	Hospital	Pediatric	Guatemala	Upper-middle income	1027	Hospital and ICU admission; inpatient LOS; mortality	Moderate
Dekker-Boersema 2019 ³⁵	January 2014	December 2016	ETAT	Hospital	Pediatric	Mozambique	Low income	4176	ED mortality	Moderate
Elbahi 2022 ³⁶	Not specified	Not specified	SATS	Hospital	Adult and pediatric	Egypt	Lower-middle income	345	ED mortality; ED LOS; hospital and ICU admission	Low
Hansoti 2016 ³⁷	March 2014	September 2014	SCREEN	Primary health care facilities	Pediatric	South Africa	Upper-middle income	3064	Waiting time (by triage category); LWBS ^d	Low
Hansoti 2017 ³⁸	March 2014	November 2016	SCREEN	Primary health care facilities	Pediatric	South Africa	Upper-middle income	5115	Waiting time; proportion who saw a professional nurse	Moderate
Khankeh 2013 ³⁹	Not specified	Not specified	Not specified	Hospital	Adult and pediatric	Iran	Lower-middle income	600	Waiting time; patient satisfaction	Moderate
Lee 2020 ⁴⁰	July 2018	January 2019	Bespoke	Hospital	Pediatric	Uganda	Low-income	10,802	Time to antibiotics	Low
Lin 2016 ⁴¹	January 2014	June 2014	CPETS	Hospital	Pediatric	China	Upper-middle income	217,009	Waiting time (overall and for urgent patients); parent satisfaction	Low
Molyneux 2006 ²⁴	January 2001	December 2003	ETAT	Hospital	Pediatric	Malawi	Low income	Not specified	Inpatient mortality	High
Mullan 2014 ⁴²	October 2009	May 2011	PATS (SATS)	Hospital	Adult and pediatric	Botswana	Lower-middle income	39,949	ED mortality; hospital admission; ICU admission; LWBS; undertriage; overtriage	Low

TABLE 1 (Continued)

First author and year	Start date	End date	Triage system ^a	Facility type ^b	Population	Country	Country income status	Total participants (n)	Effect measures	Risk of bias ^c
Robison 2012 ⁴³	November 2008	March 2010	ETAT	Hospital	Pediatric	Malawi	Low income	15,286	Early mortality (death within 2 days of admission); inpatient mortality	Moderate
Singh 2022 ⁴⁴	May 2019	April 2020	MTS	Hospital	Pediatric	India	Lower-middle income	900	Inpatient mortality (all patients and stratified by triage category); treatment initiation time (all patients and stratified by triage category) ^d	Low
Swart 2018 ⁴⁵	Not specified	Not specified	CTS (SATS)	Primary health care facilities (2)	Adult and pediatric	South Africa	Upper-middle income	720	Waiting time	Moderate
Wangara 2019 ⁴⁶	January 2015	July 2015	SATS	Hospital	Adult and pediatric	Kenya	Lower-middle income	2420	Undertriage; overtriage	Low

Abbreviations: CETEP, Clasificación Evaluación y Tratamiento de Emergencias Pediátricas; CPETS, Chinese Pediatric Emergency Triage System; ICU, intensive care unit; LOS, length of stay; LWBS, left without being seen; NOS, Newcastle-Ottawa Scale; PATS, Princess Marina Accident & Emergency Triage Scale; SCREEN, Sick Children Require Emergency Evaluation Now.

^aAcronyms for each of the triage scales utilized are defined in Table 3.

^bAll hospital-based studies were focused on triage implementation in an ED, although in many instances outcome measures included admitted patients.

^cA detailed breakdown of the risk of bias for each study, including an assessment against each of the NOS domains, is included in Appendix S3.

^dGiven the challenges of reporting data across different patient subgroups, divided into categories of urgency by differing triage systems, only subgroup data for those patients categorized as most urgent are presented here.

TABLE 2 Excluded studies.

First author and year	Triage scale	Reason for exclusion
Akadiri 2015 ⁴⁷	ETAT	Conference paper—full text unavailable
Barasa 2011 ⁴⁸	ETAT+	Conference paper—full text unavailable
Barasa 2012 ⁴⁹	ETAT+	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (ETAT+)
Becker 2015 ⁵⁰	ETAT+	Conference paper—full text unavailable
Brevidelli 2021 ⁵¹	CTAS	No reported clinical outcome or process measure
Burnett 2017 ⁵²	ETAT+	Conference paper—full text unavailable
Cartledge 2013 ⁵³	ETAT+	Conference paper—full text unavailable
Chomchai 2002 ⁵⁴	MoTS	No reported clinical outcome or process measure
Crouse 2013 ⁵⁵	ETAT	Conference paper—full text unavailable
Cruz 2020 ⁵⁶	FASTER	Conference paper—full text unavailable
Dankwa 2017 ⁵⁷	ETAT	Other (results duplicate another paper already included)
Dekker-Boersema 2017 ⁵⁸	ETAT	Conference paper—full text unavailable
Fantaye 2019 ⁵⁹	ETS	Conference paper—full text unavailable
Fateha 2001 ⁶⁰	N/A	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (ETAT+)
Hands 2018 ⁶¹	ETAT	Other (results duplicate another paper already included)
Hands 2019 ⁶²	ETAT+	Other (results duplicate another paper already included)
Hands 2020 ⁶³	ETAT+	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (ETAT+)
Hands 2021 ⁶⁴	ETAT+	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (ETAT+)
Hashemi 2013 ⁶⁵	N/A	Intervention was not the introduction of triage, or there were multiple simultaneous interventions
Hategeka 2018 ⁶⁶	ETAT+	Conference paper—full text unavailable
Hategeka 2022 ⁶⁷	ETAT+	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (ETAT+)
Irimu 2012 ⁶⁸	ETAT+	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (ETAT+)
Joshi 2020 ⁶⁹	SATS	No reported clinical outcome or process measure
Khan 2020 ⁷⁰	Bespoke (ETAT/SATS)	No reported clinical outcome or process measure
Kirollos 2018 ⁷¹	ETAT+	Conference paper—full text unavailable
Lee 2016 ⁷²	ETAT+	Conference paper—full text unavailable
Maleki 2015 ⁷³	ESI	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (a preexisting triage system was in place, and insufficient information was available to determine whether it represented a systematized approach to triage)
Mirhaghi 2015 ⁷⁴	ESI	Intervention was not the introduction of triage, or there were multiple simultaneous interventions (a preexisting triage system was in place, and insufficient information was available to determine whether it represented a systematized approach to triage)
Mitchell 2020 ²¹	IITT	No reported clinical outcome or process measure
Mitchell 2023 ⁷⁵	IITT	No reported clinical outcome or process measure
Mulindwa 2016 ⁷⁶	SATS	No reported clinical outcome or process measure (outcomes in this qualitative study were focused on staff perception of SATS and could not be readily applied or interpreted using the Donabedian framework)
Mullan 2011 ⁷⁷	PATS (SATS)	Other (results duplicate another paper already included)
Olson 2013 ⁷⁸	N/A	Intervention was not the introduction of triage, or there were multiple simultaneous interventions
Pillay 2022 ⁷⁹	Smart Triage	Conference paper—full text unavailable
Ranasinghe 2014 ⁸⁰	SATS	Conference paper—full text unavailable
Rauf 2008 ⁸¹	Not specified	Intervention was not the introduction of triage, or there were multiple simultaneous interventions
Robison 2011 ⁸²	ETAT	Other (results duplicate another paper already included)
Santosa 2015 ⁸³	ATS	No reported clinical outcome or process measure

TABLE 2 (Continued)

First author and year	Triage scale	Reason for exclusion
Twomey 2013 ⁸⁴	SATS	No reported clinical outcome or process measure
Wanefalea 2018 ²⁵	SITS	No reported clinical outcome or process measure
Wangara 2017 ⁸⁵	SATS	Other (results duplicate another paper already included)
Yuzeng 2020 ⁸⁶	PAC	Conducted in a high-income country

Abbreviations: ATS, Australasian Triage Scale; CTAS, Canadian Triage and Acuity Scale; ESI, Emergency Severity Index; ETAT, Emergency Triage Assessment and Treatment; ETAT+, Emergency Triage Assessment and Treatment Plus Admission; ETS, Emergency Triage System; FASTER, Fast Assessment and Triaging by ER Doctor; IITT, Interagency Integrated Triage Tool; MoTS, Modified Triage System; PAC, Patient Acuity Category; PATS, Princess Marina Accident & Emergency Triage Scale; SATS, South African Triage Scale; SITS, Solomon Islands Triage Scale.

TABLE 3 Triage systems featured in included studies.

Triage scale	Number of studies (n)	Description
ETAT, along with its Spanish derivative, CETEP	4	ETAT is a simple triage system developed by the WHO. ¹⁰ It allocates children to one of three categories (urgent, priority, or nonurgent) based on a combination of clinical signs and symptoms. It has been widely adopted in LMICs, particularly in Africa. Implementation of ETAT is usually accompanied by structured staff training to ensure clinicians are confident and competent to use the system. One study in this review utilized CETEP, a Spanish adaptation of ETAT. ³⁴
CPETS	1	The CPETS is a five level triage tool derived from the Canadian Triage System and Acuity Scale. It allocates children to a category of urgency based on their presenting signs and symptoms. ⁴¹
MTS	1	The MTS is a five-tier triage tool originating from the United Kingdom. The MTS comprises symptom-specific flowcharts and discriminators. Selection of a discriminator allocates the patient to the related urgency category, with a corresponding maximum waiting time.
SCREEN	2	SCREEN is a rapid assessment tool administered by laypersons. It uses six simple questions to rapidly identify and prioritize critically ill children: 1. Is your child unable to drink or breastfeed? 2. Is your child vomiting everything? 3. Has your child had convulsions during this illness? 4. Is your child lethargic or unconscious? 5. Is your child <2 months old? 6. Have you been to a clinic/hospital in the past 2 days? The outcome of this assessment is binary (ie, SCREEN positive or negative). ^{37,38}
SATS, including its predecessor, CTS, and derivative, PATS	5	SATS is a color-coded, four-tier triage instrument that uses a combination of presenting complaint and vital signs to allocate a triage category. ⁸ The triage assessment process involves calculating a "Triage Early Warning Score," which comprises the measurement of several vital signs. A modified version of the tool is used for children. PATS is a derivative of SATS, as applied at Princess Marina Hospital in Gabarone, Botswana. ⁴²
Other bespoke systems	2	One study from Uganda included in this review used a digital platform to collect clinical signs, symptoms, and vital signs to prioritize children through a combination of emergency triggers and predictive risk algorithms. ⁴⁰ Another, originating from Iran, used a "nursing triage system" that was not further defined by the authors. ³⁹
Two-step, four-level method	1	Under this system, Step 1 comprises a rapid assessment using the "pediatric assessment triangle" and basic vital signs. No specific criteria or cut points appear to have been defined. Step 2 involves allocation to one of four categories of urgency, based on descriptive criteria defined in the "Guiding Principle of Grading Pilot for Emergency Patients, 2009," published by the Ministry of Health of China. ³³

Abbreviations: CETEP, Clasificación Evaluación y Tratamiento de Emergencias Pediátricas; CPETS, Chinese Pediatric Emergency Triage System; CTS, Cape Triage Score; ETAT, Emergency Triage Assessment and Treatment; LMIC, low- and middle-income country; MTS, Manchester Triage System; PATS, Princess Marina Accident & Emergency Triage Scale; SATS, South African Triage Scale; SCREEN, Sick Children Require Emergency Evaluation Now; WHO, World Health Organization.

There were no randomized trials, and all hospital-based studies were single center. There was reasonable geographic spread, with results from five of the six WHO regions, including Africa (10 studies), Eastern Mediterranean (two studies), Southeast Asia (one study), Western Pacific (two studies), and the Americas (one study). Seven

were undertaken in upper-middle-income countries, five in lower-middle-income countries, and four in low-income country settings (Table 1).

As described in Table 3, eight different triage scales were utilized. The most widely studied was SATS⁸ (five studies) followed by

ETAT (four studies).¹⁰ All but one study reported the use of a tool specifically designed or modified for application in a LMIC context. The exception was a study by Singh and Awasthi⁴⁴ that researched the Manchester Triage Score in India.

A total of 14 effect measures (including both outcomes and process indicators) were identified, as summarized in Table 4. These are described in detail below. As anticipated, there was a significant amount of heterogeneity.

A summary of the risk-of-bias assessment for each study is provided in Table 1. Overall, nine studies were identified as being at moderate or high risk of bias. A more detailed description, including the NOS score for each domain and a summary of the critical appraisal, can be found in Appendix S3. A common issue was the lack of measures to examine and control for differences between groups, increasing the risk of confounding.

Synthesis of results related to outcomes of care

Complaints

A single study from China reported a reduction in complaints related to quality of care and crowding following triage implementation.³³ Given the paucity of data, and the risk of publication bias, the quality of evidence was rated as very low.

Hospital admission

Data were conflicting regarding the impact of triage implementation on hospital admission rates.^{34,36,42} The three studies evaluating this outcome were conducted in vastly different settings (mixed, tertiary hospitals in Botswana and Egypt and a pediatric ED in Guatemala), and this may explain the inconsistency.

Intensive care unit admission

Two studies, with data from both pediatric and adult patients, described a reduction in the rate of intensive care unit (ICU) admission.^{34,42} The authors speculate that this may reflect the timely identification and management of patients with critical illness, such that some were able to avoid ICU admission through the provision of early, effective treatment. Another study described an increase in ICU admission, but this appeared to coincide with a change in ICU admission criteria.³⁶ For this reason, there is a strong possibility of confounding.

Left without being seen

The proportion of patients leaving the ED without being seen is an important quality measure.⁸⁷ Two studies examined the impact of

triage implementation, with conflicting findings. Based on aggregate data across four study sites (primary health care clinics), Hansoti et al.³⁷ reported a statistically significant reduction in the proportion of patients who left without being seen (LWBS) by a professional nurse, but this effect was not observed across all clinics. Mullan et al.,⁴² meanwhile, reported an increase in the LWBS rate, but they speculate that this may reflect missing data in the preintervention period. Given the incongruity of the available data, the overall quality of evidence was graded as very low.

Mortality

The most common outcome measure was mortality. This was defined in a variety of ways, including deaths occurring "early" (within 48 h),⁴³ in the ED,^{35,36,42} or as an inpatient (Table 4).^{24,33,43,44} Only two of the eight studies utilizing mortality as an outcome measure included adult patients.^{36,42}

With the exception of one study,⁴² all studies utilizing this outcome measure described a reduction in mortality attributable to triage implementation.^{24,33,35,36,43,44} In most cases, this was clinically and statistically significant. For ED mortality, the maximal reported effect size was a mortality risk reduction of 47.1%.³⁶ Given the consistency of the findings across multiple studies and countries, the quality of evidence was graded as moderate.

Satisfaction

Three studies described improved patient satisfaction rates associated with the implementation of triage.^{33,39,41} A variety of instruments were used to assess satisfaction, and response rate data were not reported. For these reasons, the external generalizability of the findings is uncertain.

Synthesis of results related to process measures

ED length of stay

Two studies reporting on ED length of stay (LOS) reported a statistically significant decrease.^{33,36} Although this is a plausible effect (through a reduction in waiting time and more timely care for urgent patients), ED LOS can be influenced by a range of factors, including access block (prolonged boarding).⁸⁷ For this reason, it is unclear whether this finding is generalizable.

Inpatient LOS

The single study that investigated the impact of triage implementation on inpatient LOS found a nonsignificant reduction for critically ill children along with the broader population of admitted patients.³⁴

TABLE 4 Summary of effect measures.

Donabedian framework category	Effect measure	Specific measure	Studies (first author and year) ^a	Triage system(s)	Results (effect of triage implementation) ^b
Outcomes	Complaints	About quality	Cai 2020	Two step, four level	Cai 2020: Complaints about medical quality reduced from 21.8% to 0.6%, $p \leq 0.001$
		About crowding	Cai 2020	Two step, four level	Cai 2020: Complaints about ED crowding reduced from 74.4% to 1.9%, $p \leq 0.001$
	Hospital admission		Crouse 2016	ETAT SATS	Crouse 2016: Admission rates reduced from 8.2% to 4.3%, $p = 0.012$
			Elbaih 2022		Elbaih 2022: The proportion of patients admitted to hospital decreased from 19.1 to 17.4%, $p = 0.009^e$
	ICU admission		Mullan 2014		Mullan 2014: Admission rates increased from 41.9% (41.1%–42.7%) to 43.7% (43.1%–44.3%), $p \leq 0.05$
			Crouse 2016	ETAT SATS	Crouse 2016: Among a cohort of critically ill patients, admission rates to PICU decreased from 46.9% to 24.1%, $p = 0.002$
	LWBS		Elbaih 2022		Elbaih 2022: The proportion of patients admitted to ICU increased from 13.9 to 26.5%, $p = 0.009^e$
			Mullan 2014		Mullan 2014: The overall ICU admission rate decreased from 0.35% (0.25%–0.44%) to 0.06% (0.03%–0.09%), $p \leq 0.05$
	Deaths in ED		Hansoti 2016	SATS SCREEN	Hansoti 2016: The proportion of patients who LWBS by a professional nurse decreased from 25.8% to 18.48% post, $p \leq 0.001$
			Mullan 2014		Mullan 2014: The proportion of patients who LWBS increased from 0.01% (0.01%–0.03%) to 0.48% (0.39%–0.56%), $p \leq 0.05$
	Deaths within 48 h of admission ^a		Dekker-Boersema 2019	ETAT SATS	Dekker-Boersema 2019: Following the intervention, the mortality rate ratio was 0.55 (0.38–0.81) [$p = 0.002$], representing a 45% reduction in deaths ^c
			Elbaih 2022		Elbaih 2022: ED mortality reduced from 15.7% to 8.3%, $p = 0.043$
	Deaths among inpatients		Mullan 2014		Mullan 2014: The ED mortality rate was not impacted by the intervention, changing from 0.19% (0.12%–0.26%) to 0.19% (0.13%–0.24%) post, $p = 0.93$
			Robison 2012	ETAT	Robison 2012: Deaths occurred in 47.6/1000 admissions before the intervention compared with 37.9/1000 afterward, equating to a RR 0.8 (CI 0.67–0.93)
	Deaths among urgent and critically ill patients		Cai 2020	Two step, four level ETAT	Cai 2020: Inpatient mortality fell from 0.07% to 0.02%, $p < 0.001$
			Molyneux 2006		Molyneux 2006: Inpatient mortality decreased from “10–18% per week” to “6–8% afterwards”
	Satisfaction	Patient satisfaction	Robinson 2012		Robinson 2012: Total mortality decreased from 80.5 to 70.5 deaths per 1000 admissions (RR 0.88, 95% CI 0.78–0.98)
			Crouse 2016	ETAT MTS	Crouse 2016: In a subgroup of critically ill children, overall mortality rates decreased 52%, from 12.4% to 6.0%, $p = 0.152$
			Singh 2022		Singh 2022: Among admitted patients with immediate health care needs, mortality reduced from 55.4% to 38.2%, $p = 0.04$
			Cai 2020	Two step, four level Other	Khankeh 2013: The mean ($\pm$ SD) satisfaction score on a nationally improved survey instrument increased from 29.59 ($\pm$ 5.07) to 37.65 ($\pm$ 5.89), $p = 0.01$
	Parent satisfaction		Khankeh 2013		Cai 2020: The patient satisfaction rate improved from 73.3% to 90.0%, $p = 0.223$
			Lin 2016	CPETS	Lin 2016: Using a binary approach, parent satisfaction increased from 92.2% to 94.2%, $p < 0.001$

Direction of effect	Quality of evidence	Comments
Positive	Very low	The quality of evidence was downgraded given the limited volume of data and the risk of publication and other bias.
Conflicting data	Very low	The results of these studies demonstrated conflicting results. In the study by Mullan et al., disposition was not recorded for 12% of patients in the preintervention period. This may explain the discrepancy. On this basis, the quality of evidence has been downgraded.
Conflicting data	Low	The direction of effect was inconsistent across the studies. This may be explained by a simultaneous change in "ICU triage" criteria in the Elbahi study, which resulted in increased admissions in the interests of "improved ICU outcome [sic]."
Conflicting data	Very low	In light of conflicting evidence, the quality of evidence has been downgraded. There was also inconsistency within studies; for instance, two of four sites included by Hansoti et al. did not demonstrate a reduction in LWBS rates.
Positive	Moderate	Given the similarity between these outcome measures, they were all considered to contribute to the totality of evidence regarding the impact of triage on early mortality. The GRADE rating was upgraded to moderate on the basis of the number of studies and the consistency of effect. Although the mortality reduction demonstrated by Crouse et al. did not achieve statistical significance, this probably reflects the small sample size. The study by Mullan et al. also failed to show a mortality benefit but did not demonstrate a mortality increase. Studies utilizing inpatient mortality are potentially exposed to confounding, because the outcomes of care for admitted patients are dependent on a range of factors. However, authors utilizing this outcome measure generally maintain that, because timely recognition of critical illness is likely to enable earlier initiation of treatment, improved mortality rates should be evident among both ED and admitted patients. Despite the risk of confounding, inpatient mortality is arguably a more valid measure (compared with ED mortality) because it accounts for patients who may have survived to the point of admission but died during their hospital stay.
Positive	Very low	These ratings were downgraded given the limited volume of data and the risk of publication and reporting bias. Response rates are also unknown.

(Continues)

TABLE 4 (Continued)

Donabedian framework category	Effect measure	Specific measure	Studies (first author and year) ^a	Triage system(s)	Results (effect of triage implementation) ^b
Processes	ED LOS		Cai 2020 Elbaih 2022	Two step, four level SATS	Cai 2020: Mean $\pm$ SD ED LOS reduced from 50.5 $\pm$ 7.6 h to 20.2 $\pm$ 2.2 h, $p \leq 0.001$ Elbaih 2022: Mean ($\pm$ SD) ED LOS reduced from 183.78 ($\pm$ 93.51) min to 51.39 ($\pm$ 27.91) min, $p \leq 0.001$
	Inpatient LOS		Crouse 2016	ETAT	Crouse 2016: Inpatient LOS decreased from 4.22 days to 3.33 days, $p = 0.322$. Among a subgroup of critically ill children, inpatient LOS decreased from 7.29 days to 5.67 days, $p = 0.089$
	Overtriage		Mullan 2014 Wangara 2019	SATS	Mullan 2014: The overtriage rate, defined in the paper, fell from 52.5% (45.6%–59.4%) to 38.4% (37.5%–39.3%), $p \leq 0.001$ Wangara 2019: The overtriage rate reduced from 70.9% to 62.3%, $p \leq 0.01$
	Proportion assessed by a professional nurse ^a		Hansoti 2017	SCREEN	Hansoti 2017: The proportion of patients seen by a professional nurse increased from 6.4% to 64%, $p \leq 0.001$
	Time to antibiotics		Lee 2020	Other	Lee 2020: There was a reduction of 11 min (95% CI –16.0 to –6.0, $p < 0.001$) in the median time to antibiotics, from 51 (IQR 27.0–94.0) min to 44 (IQR 19.0–74.0) min. Children prioritized as emergency received the greatest benefit, with a reduction in the median time to antibiotics of 34 min (95% CI –9.0 to –58.0, $p < 0.001$). The proportion of patients who waited more than 1 h until antibiotics decreased by 21.4%, $p = 0.007$
	Undertriage rate		Mullan 2014 Wangara 2019	SATS	Mullan 2014: The undertriage rate fell from 46.9% (45.4%–48.3%) to 16.0% (14.6% vs 17.4%), $p \leq 0.001$ Wangara 2019: The undertriage rate changed from 3.8% to 7.8%, $p = 0.2$
	Unexpected resuscitation rate		Cai 2020	Two step, four level	Cai 2020: The unexpected resuscitation rate, defined as the “sudden requirement for rescue in patients not determined as being at risk,” decreased from 0.01% to 0.0%, $p \leq 0.001$
	Waiting time (time to assessment or consultation)	All patients	Bruijns 2008 Khankeh 2013 Lin 2016 Swart 2018	SATS Other CPETS	Bruijns 2008: There was a reduction in the mean overall waiting time from 235 (210–260) min to 146 (137–155) min, $p \leq 0.001$ Khankeh 2013: Mean waiting times decreased from 10.69 min to 8.91 min, $p = 0.01$ Lin 2016: Mean ($\pm$ SD) waiting times reduced from 41.6 ($\pm$ 15.4) min to 37.3 ($\pm$ 13.8) min, $p < 0.001$ Swart 2018: Mean ($\pm$ SD) waiting times reduced from 69.22 ($\pm$ 61.59) min to 39.86 ($\pm$ 38.42) min post, $p \leq 0.001$ ^d

Direction of effect	Quality of evidence	Comments
Positive	Low	This effect is likely to be mediated through the use of a structured process for sorting patients. It is also possible that early treatment means patients can be discharged sooner than might otherwise be the case; this may be particularly relevant for those who are categorized as urgent but do not require admission.
No effect	Very low	Given the paucity and quality of data, the level of evidence was downgraded.
Positive	Low	The quality of the evidence was considered to be low. As discussed, the use of overtriage as a performance metric carries some limitations.
Positive	Very low	This process measure is likely to reflect the unique study characteristics of the study sites, including staffing and the model of care. Although this study was thought to be at low risk of bias, the issues surrounding this indicator, and the paucity of data from other sites, led to the quality of evidence being downgraded.
Positive	Low	Only one study evaluated this process measure, but it was found to be at low risk of bias. For this reason, the quality of evidence was not adjusted.
Conflicting data	Very low	In light of conflicting evidence, and the issues associated with undertriage as a process indicator (namely, that performance is likely to be population and context dependent), the quality of evidence has been downgraded.
Positive	Very low	The quality of evidence has been downgraded given the limited amount of data and uncertainty about the data collection methods.
Positive	Moderate	As with mortality, the GRADE rating was upgraded to moderate on the basis of the number of studies and the consistency and magnitude of effect. All studies reporting data on this process measure found a positive impact. This seems to apply to both urgent patients (who stand to benefit most from triage implementation) and the broader population of ED patients. Various authors hypothesize that that this reflects the efficiency gains that come from a structured and organized approach to patient assessment and flow.

(Continues)

TABLE 4 (Continued)

Donabedian framework category	Effect measure	Specific measure	Studies (first author and year) ^a	Triage system(s)	Results (effect of triage implementation) ^b
		Urgent patients	Bruijns 2008 Hansoti 2016 Hansoti 2017 Lin 2016	SATS SCREEN CPETS	Bruijns 2008: There was a reduction in the mean waiting time for urgent (red) patients from 216 (143–289) min to 38 (28–48) min, $p \leq 0.001$ Hansoti 2016: The median waiting time for red (urgent) patients decreased from 165 min to 72 min, $p < 0.001$ Hansoti 2017: The median waiting time for red (urgent) patients decreased from 100.3 min to 4.9 min, $p < 0.001$. The proportion that were seen within 10 min increased from 6.4% to 64% Lin 2016: The median waiting time for urgent patients reduced from 3.23 (IQR 1.90–4.36) min to 2.07 (IQR 0.65–4.11) min, $p = 0.04$

Abbreviations: CPETS, Chinese Pediatric Emergency Triage System; ETAT, Emergency Triage Assessment and Treatment; ICU, intensive care unit; IQR, interquartile range; LOS, length of stay; LWBS, left without being seen; PICU, pediatric intensive care unit; RR, relative risk; SATS, South African Triage Scale; SCREEN, Sick Children Require Emergency Evaluation Now.

^aFor a summary of study characteristics, see Table 1.

^bGiven that all studies used pre-/postintervention methods, this column summarizes the effects of triage introduction (i.e., postintervention vs. preintervention).

^cThis finding likely reflects the combined effects of ETAT training and triage implementation. Further details on the limitations of this study, and the challenges in interpreting this outcome measure, are provided in Appendix S3.

^dThe reported SD for this sample is extremely broad. Relatively limited information is available on the study methods, so the veracity of this data is uncertain.

^eAlthough these data are reported as statistically significant, this was determined using a single chi-square (Fisher's exact) test applied for six different disposition outcomes.

This is a plausible effect (because timely care may have had the effect of reducing illness severity), and the reported p -values may reflect the relatively small sample size. This was the only reported outcome or process measure where there was no evidence from any study of a statistically significant positive impact.

Overtriage

Overtriage is generally defined as the proportion of patients who are allocated a high-acuity triage category but are discharged from the ED. While this definition is problematic, because not all urgent patients require admission, it is widely utilized in triage research.^{7,88}

Two studies measured overtriage rates before and after introduction of a structured triage system (SATS). In both cases, performance was compared to a retrospective cohort of patients who did not undergo any sort of formal triage (but were rather sorted using informal, local systems that lacked specific criteria). These studies demonstrated statistically significant reductions in the overtriage rate, from 52.5% to 38.4% and 70.9% to 62.3%, respectively.^{42,46}

Proportion assessed by a professional nurse

One study considered the proportion of patients who were seen by a "professional" rather than an "enrolled" nurse.³⁸ While a reduction

was evident, this finding is likely to represent the unique characteristics (such as staffing arrangements and model of care) of the study site. The external validity of these data is uncertain.

Time to antibiotics

A single study from Uganda used a bespoke, digital triage system to identify and prioritize patients with urgent care needs.⁴⁰ The primary outcome measure of the study was time to antibiotics. The study identified a reduction of 11 min (95% CI, –16.0 to –6.0 min, $p < 0.001$), from a median of 51 (interquartile range [IQR] 27.0–94.0) to 44 min (IQR 19.0–74.0 min). Children prioritized as an emergency received the greatest benefit, with a reduction in the median time to antibiotics of 34 min (95% CI –9.0 to –58.0 min, $p < 0.001$). The proportion of patients who waited more than 1 h until antibiotics decreased by 21.4% ($p = 0.007$).⁴⁰

Undertriage rate

As a marker of triage system performance, the undertriage rate shares similar limitations to the overtriage rate.^{7,88} It is generally defined as the proportion of patients who are allocated a nonurgent triage category who are subsequently admitted.^{42,46}

The same two studies that reported overtriage rates also reported undertriage rates.^{42,46} In this case, data were conflicting,

Direction of effect	Quality of evidence	Comments
no data	no data	no data

with one study demonstrating a reduction.⁴² The other showed a nonsignificant increase.⁴⁶

Unexpected resuscitation

A single study from China reported on the association between triage implementation and the “unexpected resuscitation rate.” This was defined as the “sudden requirement for rescue in patients not determined as being at risk” and decreased from 0.01% to 0.0% ($p \leq 0.001$).³³ Given a lack of clarity about study methods and generalizability, this evidence was graded as very low quality.

Waiting time

The most commonly studied process measure was time to assessment (Table 4). This reflects that a core function and priority of triage systems is to identify patients with urgent care needs and expedite their care.

While some studies focused on patients with time-critical care needs, others examined the entire population of patients presenting for EC. All demonstrated a positive and statistically significant effect. For instance, Bruijns et al.¹² considered the impact of Cape Triage Score (SATS) implementation in a South African hospital and

reported a reduction in the mean overall waiting time from 235 (210–260)min to 146 (137–155)min ($p \leq 0.001$) and a reduction in the mean waiting time for urgent patients from 216 (143–289)min to 38 (28–48)min ($p = <0.001$). Given the volume and consistency of these data, and the magnitude of effect, the quality of the evidence was graded as moderate.

DISCUSSION

Key findings

This systematic review has identified 16 studies reporting the association between triage implementation and quality of care. Data were derived from 11 countries across five geographic regions, involving a mixture of low-, lower-middle-, and upper-middle-income countries. Most studies focused exclusively on pediatric patients, with only six studies including adults.

Clinical outcomes included mortality and ICU admission, while process measures included waiting time and ED LOS. The direction of effect was positive for most indicators, including mortality, ED LOS, patient satisfaction, time to antibiotics, overtriage, and waiting time. There were conflicting data, or no evidence of association, in relation to inpatient LOS and the rates of hospital and ICU admission, LWBS, and undertriage.

Interpretation

Notwithstanding the positive and generally consistent nature of the data, there are several challenges in the interpretation of this literature. This is reflected in the fact that most studies were ranked at moderate or high risk of bias, and the majority of effect measures were assessed as providing a low or very low quality of evidence. The only two measures graded as moderate were mortality and waiting time.

Unsurprisingly, there were no randomized controlled trials (or alternate designs grounded in implementation science) among the included studies, and all shared similar limitations. Many did not adequately examine and control for difference between arms, leading to a risk of confounding. This is a common criticism of studies evaluating health systems interventions.⁸⁹ Additionally, several studies relied on retrospective medical record review to determine the characteristics of patients in a historical, preintervention cohort. The limitations of retrospective medical record review methods are well established.⁹⁰

A major issue with this literature is publication bias. Studies with negative findings, particularly those that are retrospective and observational in nature, are much less likely to have been published. This may have biased the results of this review in favor of a positive effect (i.e., inappropriate rejection of the null hypothesis). The findings should therefore be interpreted with caution.

The lack of high-quality evidence reflects the complexity of conducting research in limited resource contexts. These issues are widely acknowledged and include the challenges in accessing high-quality data (because of a lack of systematically collected medical record information and a paucity of clinical registries) as well as the limited human resources available for research (given widespread health workforce shortages and the many competing interests for busy clinicians).⁹¹ This review affirms the value of simple registry systems in LMIC ED settings, which could be used for surveillance and performance monitoring but also for audit and research purposes.⁹²

The implementation and evaluation of triage processes is a high priority for EC clinicians in LMICs.^{6,7,26,93-95} Overall, this review supports the introduction of triage in LMIC EDs, at least in relation to tools that have been designed or modified for use in resource-limited contexts and especially for pediatric patients. It would be advantageous if future studies could focus on a broader range of settings and populations, and apply more rigorous efforts to control the risk of confounding. Stepped-wedge trials would ameliorate some of these issues.⁸⁹

Limitations

The search strategy used in this review was limited to English and articles published after 1990. It is possible, therefore, that we missed a small number of articles that did not meet these criteria.

We consider that this risk is low, particularly because all included studies were published in the past two decades. Although we endeavored to contact the authors of all conference abstracts, in many cases email addresses were not provided, or we did not receive a response.

Many of the studies we identified supplemented triage implementation with structural and staffing improvements. Although we only included studies where triage was the primary intervention, excluding those that emphasized multiple concurrent changes, confounding cannot be excluded. In reality, it is difficult to implement triage without some form of training, clinical redesign process, and staffing reform.²¹ This partly reflects the power of triage to improve ED functioning, not just through the identification of patients with urgent care needs but by facilitating structural change.²²

A further potential issue with the interpretation of this literature is attribution bias. Each study implemented a single, discrete triage system, so it is impossible to determine how much of the observed effect is attributable to the function of triage (i.e., a systematized approach to sorting patients) as opposed to the specific tool that was utilized. Given that the performance of triage tools has generally been found to be consistent, and there is no compelling evidence in favor of one system over another, we consider it likely that there is some degree of "class effect."^{19,20,96-98} Moreover, as demonstrated in Table 4, data were generally consistent across all studies, even when different triage scales were utilized. Other reviews provide further information on the relative validity and reliability of different triage tools applied in LMIC ED settings.^{19,20}

As identified in other reviews, the observed performance of triage systems, and the magnitude of their impact on clinical outcomes and processes, is likely to be determined by a range of population and contextual factors.⁹⁷ For that reason, the findings summarized here will not be applicable to all EC facilities in LMICs, and the impact of triage implementation is likely to vary between sites. However, the breadth of studies included in this review would suggest that there is potential for a positive impact on quality of care across a range of EC settings and countries, especially for pediatric populations.

Implications

Notwithstanding the methodologic limitations of the review and its constituent studies, the overall body of evidence suggests that triage implementation is likely to have a range of positive impacts on LMIC ED patients and the facilities in which they seek care. This supports the intuitive value of triage, especially in contexts where demands for care frequently outstrip the available resources.

Triage has been recognized as a key priority for global EC systems improvement, and its implementation in LMICs has been strongly supported by the WHO and World Health Assembly.^{1,2,5,9} This review provides evidence that triage, as a core EC function, has the potential to improve clinical outcomes as well as some process measures. The findings will support advocacy efforts toward the

universal provision of triage, and inform future research and development activities in this area.

CONCLUSIONS

Through a systematic search and review of the literature, we identified 16 studies assessing the effects of triage implementation in LMIC EDs. We found evidence supporting an association between the introduction of triage and a reduction in deaths and waiting time, but the quality was moderate at best. The quality of evidence for other effect measures was low or very low. Although the available data support the value of triage in LMIC EDs, there is a need for further research and more robust study methods to control the risks of confounding and publication bias.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

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